Т5 $\xi \sim R(\theta, 2\theta)$, $\theta > 0$ (иначе $\theta = -1$, $[-1, -2]$?)

а) n – размер выборки

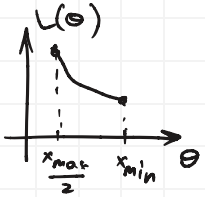
1) **метод моментов**

$$\alpha_k = \int_\theta^{2\theta} x \frac{1}{\theta} dx = \frac{x^2}{2}\Big|_\theta^{2\theta} \cdot \frac{1}{\theta} = \frac{3\theta^2}{2} \cdot \frac{1}{\theta} = \frac{3\theta}{2}$$

$$\frac{3}{2}\theta = \bar{x} \to \underline{\tilde{\theta}_1 = \frac{2}{3}\bar{x}}$$

2) **метод максимального правдоподобия**

$$L(\theta) = \prod_{i=1}^{n} p(x_i, \theta) = \prod_{i=1}^{n} \frac{1}{\theta} = \frac{1}{\theta^n}$$



$x_i \in [\theta, 2\theta] \Rightarrow x_{max} \leq 2\theta$

$x_{min} \geq \theta$

$\sup L(\theta) \Rightarrow \underline{\tilde{\theta}_2 = \frac{x_{max}}{2}}$

б)

$$M\xi = \frac{3}{2}\theta \qquad M\xi^2 = \int_\theta^{2\theta} x^2 \frac{1}{\theta} dx = \frac{x^3}{3\theta}\Big|_\theta^{2\theta} = \frac{8}{3}\theta^2 - \frac{\theta^2}{3} = \frac{7}{3}\theta^2$$

$$D\xi = \frac{7}{3}\theta^2 - \frac{9}{4}\theta^2 = \frac{\theta^2}{12}$$

1) $M\tilde{\theta}_1 = M\left[\frac{2}{3} \cdot \frac{1}{n} \sum_{i=1}^{n} x_i\right] = \frac{2}{3n} \cdot \sum_{i=1}^{n} M\xi = \theta$

$\Rightarrow \tilde{\theta}_1$ – **несмещенная**

$$D\tilde{\theta}_1 = D\left[\frac{2}{3}\frac{1}{n}\sum_{i=1}^{n} x_i\right] = \frac{4}{9n^2}\cdot nD\xi = \frac{\theta^2}{27n} \xrightarrow[n\to\infty]{} 0$$

$\Rightarrow \tilde{\theta}_1$ – состоятельна по дост. условию

2) $M\tilde{\theta}_2 = \frac{1}{2}Mx_{max} = \frac{1}{2}\int_{-\infty}^{+\infty} x\,d\Phi(x) = \frac{1}{2}\int_{\theta}^{2\theta} x\varphi(x)dx$

$\Phi(x) = (F(x))^n$; $\varphi(x) = \Phi'(x) = np(x)(F(x))^{n-1} =$

$= n\frac{1}{\theta}\cdot\left(\frac{x}{\theta} - 1\right)^{n-1} = \frac{n}{\theta^n}(x-\theta)^{n-1}$ $\{(\theta, 2\theta)\}$

$\frac{1}{2}Mx_{max} = \frac{1}{2}\int_{\theta}^{2\theta}\frac{n}{\theta^n}(x-\theta)^{n-1}x\,dx = \frac{n}{2\theta^n}\int_0^{\theta} t^{n-1}(t+\theta)dt =$

$= \frac{n}{2\theta^n}\left(\frac{\theta^{n+1}}{n+1} + \frac{\theta^{n+1}}{n}\right) = \theta\frac{2n+1}{2n+2} = M\tilde{\theta}_2$

$\Rightarrow \tilde{\theta}_2$ – смещенная

Исправим $\tilde{\theta}_2' = \frac{2n+2}{2n+1}\tilde{\theta}_2 = \frac{n+1}{2n+1}x_{max}\cdot\frac{2}{2}$

$\Rightarrow \tilde{\theta}_2'$ – несмещённая оценка

$M\tilde{\theta}_2^2 = \frac{1}{4}Mx_{max}^2 = \frac{1}{4}\frac{n}{\theta^n}\int_{\theta}^{2\theta}(x-\theta)^{n-1}x^2dx =$

$= \frac{n}{4\theta^n}\int_0^{\theta} t^{n-1}(t^2+2t\theta+\theta^2)dt = \frac{n}{4\theta^n}\left(\frac{\theta^{n+2}}{n+2} + 2\frac{\theta^{n+2}}{n+1} + \frac{\theta^{n+2}}{n}\right)$

$= \frac{\theta^2}{4}n\left(\frac{n^2+n+2n^2+4n+n^2+3n+2}{n(n+1)(n+2)}\right) = \frac{\theta^2}{4}\cdot\frac{4n^2+8n+2}{(n+1)(n+2)}$

$= \theta^2\frac{2n^2+4n+1}{2(n+1)(n+2)}$

$$D\tilde{\theta}_2 = \theta^2 \frac{2n^2+4n+1}{2(n+1)(n+2)} - \theta^2 \frac{(2n+1)^2}{4(n+1)^2} = \theta^2 \left(\frac{(2n^2+4n+1)\cdot(2n+2)}{4(n+1)^2(n+2)} - \right.$$

$$\left. - \frac{(4n^2+4n+1)(n+2)}{4(n+1)^2(n+2)} \right) = \theta^2 \left(\frac{4n^3+12n^2+10n+2 - 4n^3 - 12n^2 - 9n - 2}{4(n+1)^2(n+2)} \right) =$$

$$= \frac{\theta^2}{4} \cdot \frac{n}{(n+1)^2(n+2)} \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_2 - \text{состоятельна}$$

$$D\tilde{\theta}_2' = \frac{4(n+1)^2}{(2n+1)^2} \cdot \frac{\theta^2}{4} \frac{n}{(n+1)^2(n+2)} = \theta^2 \frac{n}{(n+2)(2n+1)^2} \xrightarrow[n\to\infty]{} 0$$

$$\Rightarrow \tilde{\theta}_2' - \text{состоятельна}$$

c) сравнение эффективности оценок

$$\tilde{\theta}_1 = \frac{2}{3}\bar{x} \quad \text{и} \quad \tilde{\theta}_2' = \frac{n+1}{2n+1} x_{max}$$

$$D\tilde{\theta}_1 = \frac{\theta^2}{27n} \quad \text{и} \quad D\tilde{\theta}_2' = \frac{\theta^2 n}{(n+2)(2n+1)^2}$$

$$\frac{1}{27n} \gtrless \frac{n}{(n+2)(2n+1)^2}$$

$$(n+2)(4n^2+4n+1) \gtrless 27n^2$$

$$4n^3+12n^2+9n+2 \gtrless 27n^2$$

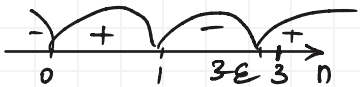
$$4n^3-15n^2+9n+2 \gtrless 0$$

$$(n-1)(4n^2-11n-2) \gtrless 0$$

$$D = 153 = (3\sqrt{17})^2$$

$$n_1 = \frac{11+3\sqrt{17}}{8} = 3-\varepsilon,$$

$$n_2 = \frac{11-3\sqrt{17}}{8} \simeq 0 \quad \varepsilon > 0, \ \varepsilon \sim 0$$



Эффективность:

при $n \in [1,2]$: $D\tilde{\theta}_1 \leq D\tilde{\theta}_2'$

при $n \geq 3$: $D\tilde{\theta}_1 > D\tilde{\theta}_2'$

при $n \in [1,2]$: $\tilde{\theta}_1$ эфф. $\tilde{\theta}_2'$

при $n \geq 3$: $\tilde{\theta}_2'$ эфф. $\tilde{\theta}_1$

d) У нас $\xi \sim R(\theta, 2\theta)$

$\theta \le x_i \le 2\theta$

$1 \le \frac{x_i}{\theta} \le 2$

$0 \le \frac{x_i}{\theta} - 1 \le 1 \quad (\forall i)$

Удобно взять $f(\vec{x_n}, h) = \frac{x_{max}}{\theta} - 1$,

т. к. $P(x) = (F(x))^n$.

$$P\left(\frac{x_{max}}{\theta} - 1 < t\right) = P(x_{max} < \theta(1+t)) =$$

$$= (F(\theta(1+t)))^n; \quad F(x) = \frac{x-\theta}{\theta}$$

$$F(\theta(1+t)) = t \Rightarrow P\left(\frac{x_{max}}{\theta} - 1 < t\right) = t^n, \quad t \in [0,1]$$

$t_1: F(t_1) = \frac{1-\beta}{2}$

$t_2: F(t_2) = \frac{1+\beta}{2}$

$$\Rightarrow \sqrt[n]{\frac{1-\beta}{2}} < \frac{x_{max}}{\theta} - 1 < \sqrt[n]{\frac{1+\beta}{2}}$$

$P(\dots) = \beta$

$$\Rightarrow \frac{x_{max}}{\sqrt[n]{\frac{1+\beta}{2}} + 1} < \theta < \frac{x_{max}}{\sqrt[n]{\frac{1-\beta}{2}} + 1}$$

e) $\tilde{\theta}_1 = \frac{2}{3}\bar{x}$ $\quad \sqrt{n}(\bar{x} - M\xi) \rightsquigarrow N(0, \frac{\theta^2}{12})$

$$\sqrt{n}(\tilde{\theta}_1 - \theta) = \frac{2}{3}\sqrt{n}(\bar{x} - M\xi) \rightsquigarrow N(0, \frac{4}{9}\cdot\frac{\theta^2}{12})$$

$$\sqrt{n}(\tilde{\theta}_1 - \theta) \rightsquigarrow N(0, \frac{\theta^2}{27})$$

$$\frac{\tilde{\theta}_1 - \theta}{\frac{\theta}{\sqrt{27n}}} \rightsquigarrow N(0,1)$$

$t_1 = u_{\frac{1-\beta}{2}}$

$t_2 = u_{\frac{1+\beta}{2}}$

$$P\left(t_1 < \frac{\tilde{\theta}_1 - \theta}{\frac{\theta}{\sqrt{27n}}} < t_2\right) = \beta$$

$$\tilde{\theta}_1 - t_2\frac{\theta}{\sqrt{27n}} < \theta < \tilde{\theta}_1 - t_1\frac{\theta}{\sqrt{27n}}$$

$h \to \infty$: $\theta \to \tilde{\theta}_1 \Rightarrow \tilde{\theta}_1 - t_2\frac{\tilde{\theta}_1}{\sqrt{27n}} < \theta < \tilde{\theta}_1 - t_1\frac{\tilde{\theta}_1}{\sqrt{27n}}$

f, g, h

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import scipy.stats as st
import seaborn as sns
import math as mt
from skimage.io import imread, imsave

np.random.seed(777)
```

In [2]:
```
n = 100
beta = 0.95

theta = 25 # <- захотели
X = st.uniform(loc=theta, scale=2*theta-theta).rvs(size=n)
```

In [3]:
```
Mx = np.mean(X)
x_max = np.max(X)

theta1 = 2/3 * Mx                       # <- оценка, полученная ОММ
theta2 = (n + 1) * x_max / (2*n + 1)    # <- исправленная оценка, полученная ОМП
```

f) Для сгенерированной выборки размера $n = 100$ и $\theta = 25$ вычислим доверительные интервалы

In [4]:
```
# Точный метод

d_left = x_max/(np.pow(1 + beta, 1/n) + 1)
d_right = x_max/(np.pow(1 - beta, 1/n) + 1)
accurate_l = d_right - d_left

print(f"Точный доверительный интервал для theta: {d_left:.5f} ≲ theta ≲
{d_right:.5f}")
print(f"Длина интервала l \u2248 {accurate_l:.5f}")
```

```
Точный доверительный интервал для theta: 24.91103 ≲ theta ≲ 25.36885
Длина интервала l ≈ 0.45782
```

In [5]:
```
# Асимптотический метод

u = st.norm.ppf((1 + beta)/2)

d_left = theta1 - u * theta1 / np.sqrt(27*n)
d_right = theta1 + u * theta1 / np.sqrt(27*n)
asymptotic_l = d_right - d_left

print(f"Асимптотический доверительный интервал для theta: {d_left:.5f} ≲ theta ≲
{d_right:.5f}")
print(f"Длина интервала l ≈ {asymptotic_l:.5f}")
```

```
Асимптотический доверительный интервал для theta: 23.82722 ≲ theta ≲ 25.69518
Длина интервала l ≈ 1.86796
```

g) Численно построить бутстраповский интервал

In [6]:
```
N = 1000

def omp_bootstrap(x, N):
    arr = list()
    n = len(x)

    for _ in range(N):
        sample = np.random.choice(x, size=n, replace=True)
        arr += [(n+1) * np.max(sample) / (2*n + 1) - theta2]

    return sorted(np.array(arr))

bootstrap_array = omp_bootstrap(X, N)

t1 = int((1 - beta)/2 * N - 1)
t2 = int((1 + beta)/2 * N - 1)

bootstrap_left = theta1 - bootstrap_array[t2]
bootstrap_right = theta1 - bootstrap_array[t1]
bootstrap_l_omp = bootstrap_right - bootstrap_left

print(f"Непараметрический бутстраповский доверительный интервал для theta (ОМП):
{bootstrap_left:.5f} ≲ theta ≲ {bootstrap_right:.5f}")
print(f"Длина интервала l \u2248 {bootstrap_l_omp:.5f}")
```

```
Непараметрический бутстраповский доверительный интервал для theta (ОМП): 24.76120 ≲
theta ≲ 25.19219
Длина интервала l ≈ 0.43099
```

In [7]:
```
N = 1000

def omm_bootstrap(x, N):
    arr = list()
    n = len(x)

    for _ in range(N):
        sample = np.random.choice(x, size=n, replace=True)
        arr += [2/3 * np.mean(sample) - theta1]

    return sorted(np.array(arr))

bootstrap_array = omm_bootstrap(X, N)

t1 = int((1 - beta)/2 * N - 1)
t2 = int((1 + beta)/2 * N - 1)

bootstrap_left = theta1 - bootstrap_array[t2]
bootstrap_right = theta1 - bootstrap_array[t1]
bootstrap_l_omm = bootstrap_right - bootstrap_left

print(f"Непараметрический бутстраповский доверительный интервал для theta (ОММ):
{bootstrap_left:.5f} ≲ theta ≲ {bootstrap_right:.5f}")
print(f"Длина интервала l \u2248 {bootstrap_l_omm:.5f}")
```

```
Непараметрический бутстраповский доверительный интервал для theta (ОММ): 23.86801 ≲
theta ≲ 25.65124
Длина интервала l ≈ 1.78323
```

h) Сравнить все интервалы

In [8]:
```
from tabulate import tabulate

lengths = {
    "Точный": accurate_l,
    "Асимптотический": asymptotic_l,
    "Бутстрап ОММ": bootstrap_l_omm,
    "Бутстрап ОМП": bootstrap_l_omp
}

sorted_lengths = sorted([(value, name) for name, value in lengths.items()])

table_data = []
for i, (length, name) in enumerate(sorted_lengths, start=1):
    table_data.append([i, name, np.round(length, 3)])

print(tabulate(table_data, headers=["№", "Метод", "Длина (l)"],
tablefmt="fancy_outline"))
```

№	Метод	Длина (l)
1	Бутстрап ОМП	0.431
2	Точный	0.458
3	Бутстрап ОММ	1.783
4	Асимптотический	1.868